

QuantumShield Benchmark Results

Post-Quantum Cryptography Performance Evaluation
KEMTLS vs Classical TLS · ML-DSA-65 vs RSA-2048

1. Handshake Latency

100 iterations, loopback interface. KEMTLS uses ML-KEM-768 for key encapsulation; classical TLS uses ECDH P-256 + RSA-2048.

Protocol	Mean (ms)	Median (ms)	Stdev (ms)	Min (ms)	Max (ms)
TLS	1.38	1.24	0.52	0.84	7.54
KEMTLS	1.08	0.95	0.46	0.51	4.25

✓ KEMTLS mean handshake: **1.08 ms** vs TLS **1.38 ms** — **21.6% faster** on average.

2. Token Generation Latency

JWT signing benchmark — 100 iterations per algorithm.

Algorithm	Mean (ms)	Median (ms)	Stdev (ms)	Min (ms)	Max (ms)
RSA-2048	0.61	0.52	0.29	0.30	2.35
ML-DSA-65	0.76	0.61	0.50	0.21	5.60

3. Token Verification Latency

JWT signature verification — 100 iterations per algorithm.

Algorithm	Mean (ms)	Median (ms)	Stdev (ms)	Min (ms)	Max (ms)
RSA-2048	0.097	0.083	0.053	0.058	0.70
ML-DSA-65	0.214	0.201	0.065	0.141	0.57

4. Key and Signature Sizes

Artifact	TLS / RSA-2048 (bytes)	KEMTLS / ML-DSA-65 (bytes)
Public Key	294	1,184
Signature	256	3,309
Token Signature	256	3,309

Note: Post-quantum signatures are larger by design, trading bandwidth for quantum resistance. Key sizes are acceptable for modern TLS record sizes.

5. End-to-End Auth & Userinfo Latency

Full OIDC round-trip including handshake, 100 iterations.

Endpoint	Protocol	Mean (ms)	Median (ms)	Min (ms)	Max (ms)
Auth	TLS	2.29	2.02	1.22	59.63

Auth	KEMTLS	2.19	1.98	0.97	15.52
Userinfo	TLS	0.098	0.082	0.059	0.83
Userinfo	KEMTLS	0.217	0.198	0.145	0.93

All measurements taken on localhost loopback. Numbers match benchmark_results.csv and BENCHMARKS.md. Environment:
Python 3.10, liboqs 0.10.0, Ubuntu 24.